PREVIOUSLY · V2 · FOUR WEEKS AGO

I posted a 57.9% result. The number I can defend is 22.3%.

~~57.9%~~ **22.3%**

CORRECTED

The baseline was a 50% cut-off. At a 1.1% fraud rate, that is barely better than approve-all.

WHAT V2 TESTED

Does the result survive harder operating conditions?

6
tests with kill criteria written first

3
survived · one fragile · two killed

44.25%
under a simulated 2% review cap

Review capacity, label delay, retraining, drift, block-rate disparity, calibration. Kill criteria written before each test ran.

Source: reports/v2_falsification.md

THEN I ASKED A DIFFERENT QUESTION

Was the baseline fair?

0.50
the V1 baseline cut-off

1.1%
the fraud base rate

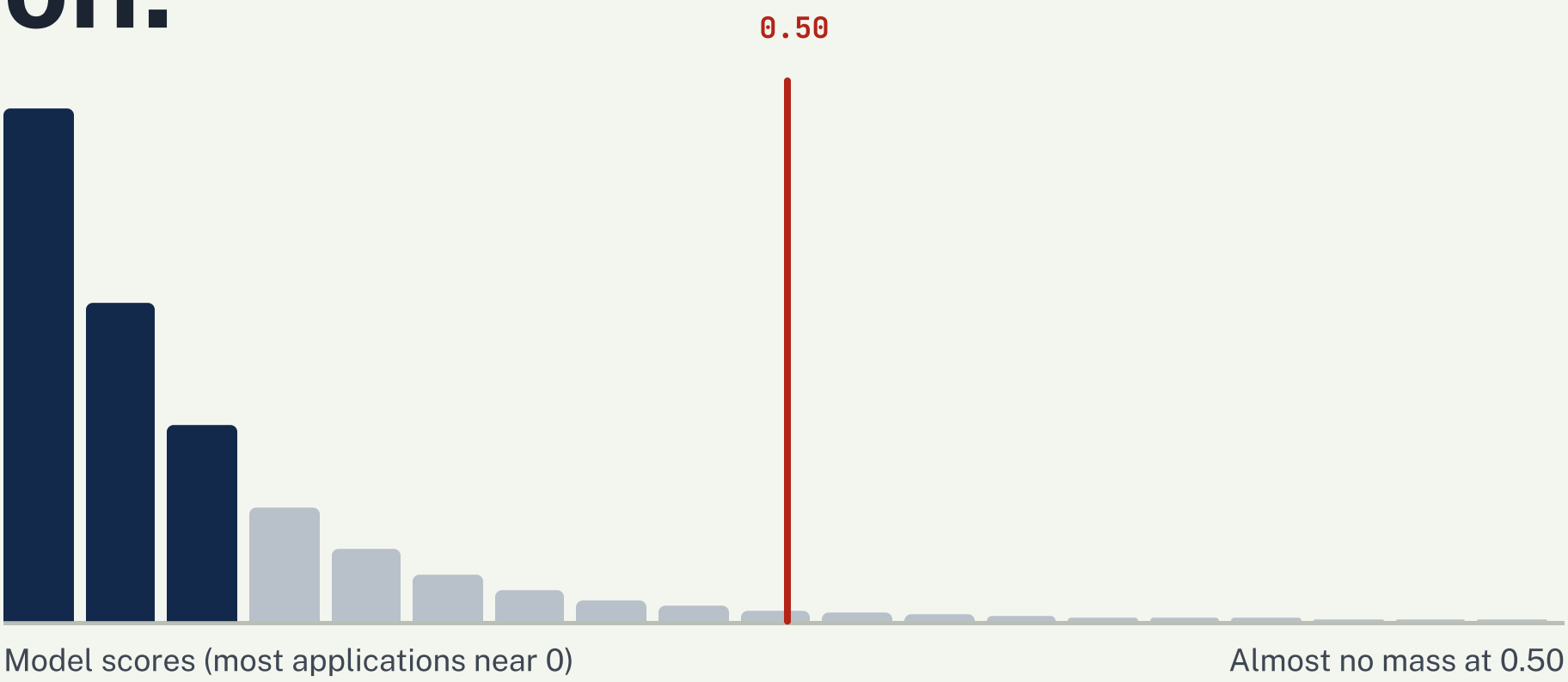
≈ 0
applications that reach 0.50

V2 tested harder conditions. It did not test whether the comparison was fair. This update does.

Source: reports/ablation.md

THE GAP

The baseline was a 50% cut-off.

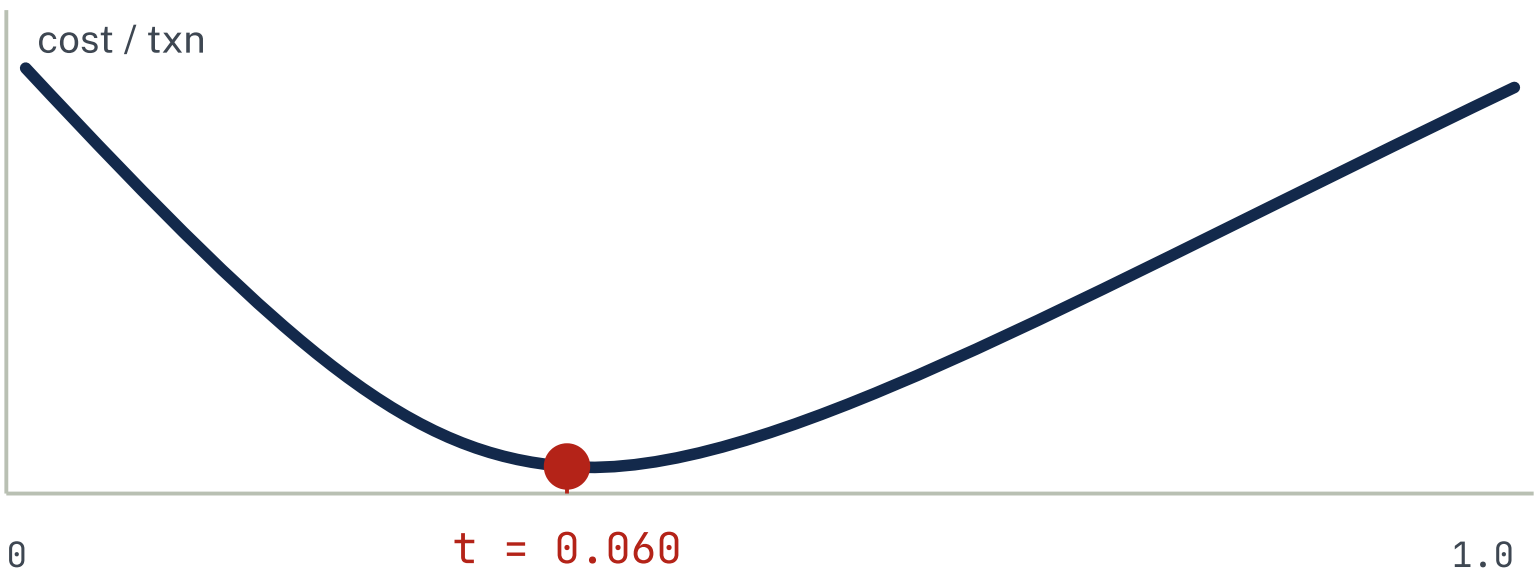


At a 1.1% base rate, almost no application reaches 0.50. The 57.9% was mostly threshold placement.

Source: reports/ablation.md

THE FIX

So I rebuilt the comparison.



A strong simple baseline: a cost-optimal threshold tuned on validation and applied unchanged to test.

Source: src/evaluation/baselines.py

THE CORRECTED HEADLINE

57.9% → 22.3%



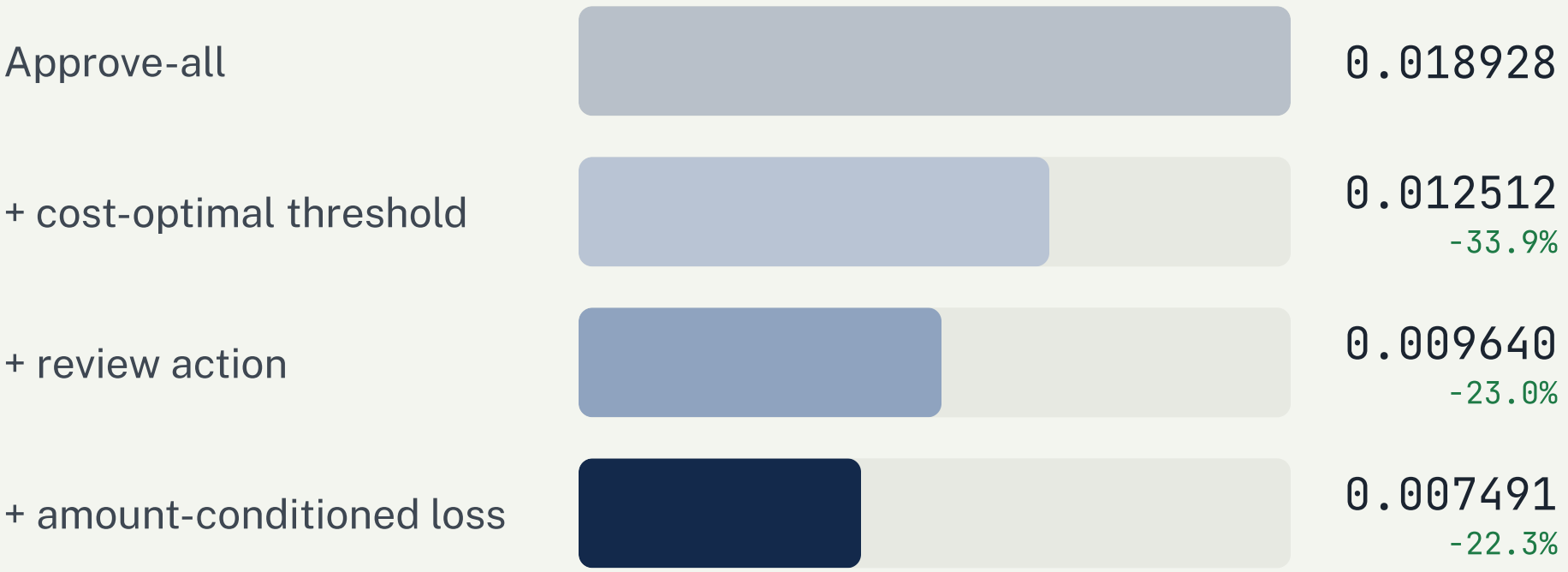
V1 · vs a 50% cut-off
95% CI 53.95–62.13

V2.1 · vs the best cost-optimal threshold
95% CI 19.71–24.95

Smaller. Defensible. The 22.3% is above the 5% effect-size bar, and the 95% CI excludes zero.

WHERE THE GAIN COMES FROM

Three steps, not one.



The steps are sequential, not additive. Threshold placement, then the review action, then amount conditioning.

Source: reports/ablation.md

WHAT THIS DOES TO V2

The six verdicts are unchanged. The headline shrinks.

HYPOTHESIS	VERDICT	KEY RESULT
Review capacity	SURVIVED	44.25% under a 2% cap
Longer label delay	SURVIVED	58.08%*
Rolling retraining	FRAGILE	LR selected; 4.25% margin
Feature drift	SURVIVED	9/28; mostly low-importance
Group disparity	KILLED	2.3 pts vs a 5-pt bar
Stratified calibration	KILLED	bias ~2.1%

The model, cost matrix, split, and policy are unchanged. So are the caveats.

*H2 uses a separate 2-month split, so it is not a like-for-like delay comparison. H5 tested block-rate disparity only; review-rate disparity was larger and not pre-registered.

Source: reports/v2_falsification.md

THE LESSON

Robustness testing is not only about harder conditions.



It is also about a fair comparison. I stress-tested the first four weeks ago and missed the second. The smaller number is the one I can defend.

V2.1 IS LIVE

An updated case study, an ablation report, and the same six verdicts.

76	227,491	6
unit and integration tests	rows in the frozen V1 test window	tests with kill criteria written first

Repo: github.com/kira-ml/delayed-label-fraud-decisioning

Demo: delayed-label-fraud-decisioning-gefp9s9mbkfdyzhhvescdm.streamlit.app

One-pager: V2.1 update in this folder

I'm a third-year ML student. I built a decision system for a delayed-feedback problem — and this is what correcting my own result looked like.